CHAPTER 5

SECONDARY ACTIVITIES

CHAPTER ESSENCE AT A GLANCE:
Examine the transformation of raw materials into finished commodities through manufacturing, factors determining industrial location, classification of industries by scale, raw materials, outputs, and ownership, footloose industries, and high-tech industrial complexes.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Characteristics of Modern Large-Scale Manufacturing:** Characterized by rigorous division of labour, specialized skills, continuous mechanization, automation, technological innovation, complex executive hierarchy, and geographical clustering.

• **Determinants of Industrial Location:** Industrial plants locate at points of minimum cost where raw material availability, cheap power, accessible markets, transport links, skilled labour, and government policies intersect.

• **Classification by Scale and Raw Materials:** Classified by scale into household or cottage, small-scale, and large-scale manufacturing; classified by raw materials into agro-based, mineral-based, chemical-based, forest-based, and animal-based industries.

• **Footloose Industries and High-Technology Parks:** Footloose industries do not rely on raw material weight loss and can locate flexibly along road networks. High-tech manufacturing relies on research, technopolies, and innovation clusters.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$\text{Manufacturing Industry Value Added} = \text{Value of Finished Product} - \text{Cost of Raw Materials}$$

Value Addition in Secondary Manufacturing

$$\text{Agglomeration Economies} \rightarrow \text{Clustering of Enterprises} \rightarrow \text{Shared Infrastructure and Labour}$$

Cost Reduction through Industrial Agglomeration

$$\text{Footloose Industries} \iff \text{Component Assembly} + \text{Negligible Weight Loss} + \text{Zero Specific Raw Material Pull}$$

Flexible Locational Freedom of Footloose Units

$$\text{Industrial Weberian Location Index} \iff \frac{\text{Weight of Localised Raw Materials}}{\text{Weight of Finished Manufactured Product}}$$

Material Index Governing Industrial Spatial Pull

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Footloose industries do not cause heavy environmental pollution; they are clean assembly operations using prefabricated components that produce small-volume outputs.

• **Alert:** Petrochemical industries are classified as chemical-based and mineral-oil-based, never agro-based.

• **Alert:** Cottage or household industries rely entirely on family labour and local raw materials, producing everyday consumer items with zero power-driven factory machinery.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Explain any three factors that govern the location of manufacturing industries. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **1. Access to Raw Materials:** Industries using heavy, bulky, or weight-losing raw materials (such as iron ore or sugarcane) locate near the raw material sources. **[1 Mark]**

• **2. Access to Market:** Industries manufacturing perishable goods or heavy finished items locate close to large consumer markets to minimize freight overheads. **[1 Mark]**

• **3. Access to Power and Transport:** Reliable electrical power and well-connected rail and road networks are essential to assemble inputs and dispatch products. **[1 Mark]**

2. Distinguish between household or cottage industry and small-scale industry. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Cottage Industry:** Smallest manufacturing unit operated by family artisans in their homes using simple tools and local raw materials; goods are made for local consumption with minimal capital. **[1.5 Marks]**

• **Small-Scale Industry:** Uses power-driven machinery, hired semi-skilled labour, and workshops outside the home; consumes raw materials from outside and caters to broader urban markets. **[1.5 Marks]**

3. What are footloose industries? State any two essential features of such industries. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition:** Manufacturing enterprises that can locate in a wide variety of places without being tied to specific raw material sources. **[1 Mark]**

• **Key Features:** (a) They depend on component parts produced elsewhere rather than raw natural ores; (b) They produce in small quantities with non-polluting operations and rely heavily on road transport networks. **[2 Marks]**

4. Differentiate between agro-based industries and mineral-based industries with two examples of each. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Agro-Based Industries:** Involve processing raw materials harvested from agricultural fields and rural farms into manufactured consumer products. Examples include sugar industry, cotton textiles, and food canning. **[2 Marks]**

• **Mineral-Based Industries:** Use metallic and non-metallic mineral ores extracted from the earth crust as their primary industrial inputs. Examples include iron and steel manufacturing, copper smelting, and cement production. **[2 Marks]**

5. Why is the iron and steel industry called a basic industry and a heavy industry? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Basic Industry:** Its finished steel outputs provide the primary raw material and machinery for all other manufacturing industries (railways, automobiles, tools). **[1.5 Marks]**

• **Heavy Industry:** Uses bulky, weight-losing raw materials (iron ore, coking coal, limestone) and produces heavy, bulky machinery requiring heavy-duty freight transport. **[1.5 Marks]**

6. Explain the concept of agglomeration economies in industrial geography. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Definition:** Cost savings and economic benefits that accrue to manufacturing firms when they cluster geographically close to one another in industrial hubs. **[1 Mark]**

• **Shared Benefits:** Enables firms to share common infrastructure, specialized labour pools, financial services, and transportation links, lowering per-unit production costs. **[1 Mark]**

7. What are high-technology industries? Explain how technopolies and science parks foster high-tech manufacturing. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **High-Tech Industries:** The latest generation of manufacturing that integrates advanced scientific research, development, computer-aided design, robotics, and biotechnology. **[1.5 Marks]**

• **Technopolies and Science Parks:** Planned regional clusters where high-tech companies, university research laboratories, and venture capital assemble in pristine campus environments. **[1.5 Marks]**

• **Global Exemplars:** Silicon Valley near San Francisco, Route 128 near Boston, and electronic cities like Bengaluru in India. **[1 Mark]**